

## Objective

- Clean and explore the dataset
- Visualize unemployment trends
- Study the impact of COVID-19
- Detect seasonal or long-term patterns
- Present insights useful for economic and social policy decisions

```
# Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Load the dataset
df = pd.read_csv(r'/content/Unemployment in India.csv')
```

df

|     | Region         | Date       | Frequency | Estimated Unemployment Rate (%) | Estimated Employed | Estimated Labour Participation Rate (%) | Area  |
|-----|----------------|------------|-----------|---------------------------------|--------------------|-----------------------------------------|-------|
| 0   | Andhra Pradesh | 31-05-2019 | Monthly   | 3.65                            | 11999139.0         | 43.24                                   | Rural |
| 1   | Andhra Pradesh | 30-06-2019 | Monthly   | 3.05                            | 11755881.0         | 42.05                                   | Rural |
| 2   | Andhra Pradesh | 31-07-2019 | Monthly   | 3.75                            | 12086707.0         | 43.50                                   | Rural |
| 3   | Andhra Pradesh | 31-08-2019 | Monthly   | 3.32                            | 12285693.0         | 43.97                                   | Rural |
| 4   | Andhra Pradesh | 30-09-2019 | Monthly   | 5.17                            | 12256762.0         | 44.68                                   | Rural |
| ... | ...            | ...        | ...       | ...                             | ...                | ...                                     | ...   |
| 763 | NaN            | NaN        | NaN       | NaN                             | NaN                | NaN                                     | NaN   |
| 764 | NaN            | NaN        | NaN       | NaN                             | NaN                | NaN                                     | NaN   |
| 765 | NaN            | NaN        | NaN       | NaN                             | NaN                | NaN                                     | NaN   |
| 766 | NaN            | NaN        | NaN       | NaN                             | NaN                | NaN                                     | NaN   |
| 767 | NaN            | NaN        | NaN       | NaN                             | NaN                | NaN                                     | NaN   |

768 rows × 7 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

## Explore the dataset

```
# display the first 5 rows
print(df.head())
```

```

      Region      Date  Frequency  Estimated Unemployment Rate (%) \
0  Andhra Pradesh  31-05-2019   Monthly                3.65
1  Andhra Pradesh  30-06-2019   Monthly                3.05
2  Andhra Pradesh  31-07-2019   Monthly                3.75
3  Andhra Pradesh  31-08-2019   Monthly                3.32
4  Andhra Pradesh  30-09-2019   Monthly                5.17

      Estimated Employed  Estimated Labour Participation Rate (%)  Area
0          11999139.0                43.24  Rural
1          11755881.0                42.05  Rural
2          12086707.0                43.50  Rural
3          12285693.0                43.97  Rural
4          12256762.0                44.68  Rural
```

```
# dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 7 columns):
```

```
#
```

| # | Column                                  | Non-Null Count | Dtype   |
|---|-----------------------------------------|----------------|---------|
| 0 | Region                                  | 740 non-null   | object  |
| 1 | Date                                    | 740 non-null   | object  |
| 2 | Frequency                               | 740 non-null   | object  |
| 3 | Estimated Unemployment Rate (%)         | 740 non-null   | float64 |
| 4 | Estimated Employed                      | 740 non-null   | float64 |
| 5 | Estimated Labour Participation Rate (%) | 740 non-null   | float64 |
| 6 | Area                                    | 740 non-null   | object  |

```

dtypes: float64(3), object(4)
memory usage: 42.1+ KB

```

```
#check missing value
df.isnull().sum()
```

```
0
```

| Region                                  | 28 |
|-----------------------------------------|----|
| Date                                    | 28 |
| Frequency                               | 28 |
| Estimated Unemployment Rate (%)         | 28 |
| Estimated Employed                      | 28 |
| Estimated Labour Participation Rate (%) | 28 |
| Area                                    | 28 |

```
dtype: int64
```

```
#statistical summary
df.describe()
```

|       | Estimated Unemployment Rate (%) | Estimated Employed | Estimated Labour Participation Rate (%) |
|-------|---------------------------------|--------------------|-----------------------------------------|
| count | 740.000000                      | 7.400000e+02       | 740.000000                              |
| mean  | 11.787946                       | 7.204460e+06       | 42.630122                               |
| std   | 10.721298                       | 8.087988e+06       | 8.111094                                |
| min   | 0.000000                        | 4.942000e+04       | 13.330000                               |
| 25%   | 4.657500                        | 1.190404e+06       | 38.062500                               |
| 50%   | 8.350000                        | 4.744178e+06       | 41.160000                               |
| 75%   | 15.887500                       | 1.127549e+07       | 45.505000                               |
| max   | 76.740000                       | 4.577751e+07       | 72.570000                               |

## ▼ Data Cleaning

```
# Rename columns for easier use
df.column = df.columns.str.strip()
```

```
/tmp/ipykernel_6556/1337854070.py:2: UserWarning: Pandas doesn't allow columns to be created via a new attribute name - see
df.column = df.columns.str.strip()
```

```
df.head()
```

|   | Region         | Date       | Frequency | Estimated Unemployment Rate (%) | Estimated Employed | Estimated Labour Participation Rate (%) | Area  |
|---|----------------|------------|-----------|---------------------------------|--------------------|-----------------------------------------|-------|
| 0 | Andhra Pradesh | 31-05-2019 | Monthly   | 3.65                            | 11999139.0         | 43.24                                   | Rural |
| 1 | Andhra Pradesh | 30-06-2019 | Monthly   | 3.05                            | 11755881.0         | 42.05                                   | Rural |
| 2 | Andhra Pradesh | 31-07-2019 | Monthly   | 3.75                            | 12086707.0         | 43.50                                   | Rural |
| 3 | Andhra Pradesh | 31-08-2019 | Monthly   | 3.32                            | 12285693.0         | 43.97                                   | Rural |
| 4 | Andhra Pradesh | 30-09-2019 | Monthly   | 5.17                            | 12256762.0         | 44.68                                   | Rural |

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
print(df.columns)
```

```
Index(['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)',
      'Estimated Employed', 'Estimated Labour Participation Rate (%)',
      'Area'],
      dtype='object')
```

```
# Remove missing values
df.dropna(inplace=True)
```

```
#check duplicate
df.drop_duplicates(inplace=True)
```

```
df.head()
```

|   | Region         | Date       | Frequency | Estimated Unemployment Rate (%) | Estimated Employed | Estimated Labour Participation Rate (%) | Area  |
|---|----------------|------------|-----------|---------------------------------|--------------------|-----------------------------------------|-------|
| 0 | Andhra Pradesh | 31-05-2019 | Monthly   | 3.65                            | 11999139.0         | 43.24                                   | Rural |
| 1 | Andhra Pradesh | 30-06-2019 | Monthly   | 3.05                            | 11755881.0         | 42.05                                   | Rural |
| 2 | Andhra Pradesh | 31-07-2019 | Monthly   | 3.75                            | 12086707.0         | 43.50                                   | Rural |
| 3 | Andhra Pradesh | 31-08-2019 | Monthly   | 3.32                            | 12285693.0         | 43.97                                   | Rural |
| 4 | Andhra Pradesh | 30-09-2019 | Monthly   | 5.17                            | 12256762.0         | 44.68                                   | Rural |

Next steps:

[Generate code with df](#)[New interactive sheet](#)

## 5. Understanding Key Columns

- Typical columns in unemployment datasets:
- Region
- Date
- Estimated Unemployment Rate (%)
- Estimated Employed
- Estimated Labour Participation Rate (%)

```
df.columns = df.columns.str.strip()
```

```
print(df.columns)
```

```
Index(['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)',
      'Estimated Employed', 'Estimated Labour Participation Rate (%)',
      'Area'],
      dtype='object')
```

```
print('Date' in df.columns)
```

```
True
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

```
/tmp/ipykernel_6556/2394721818.py:1: UserWarning: Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was sp
df['Date'] = pd.to_datetime(df['Date'])
```

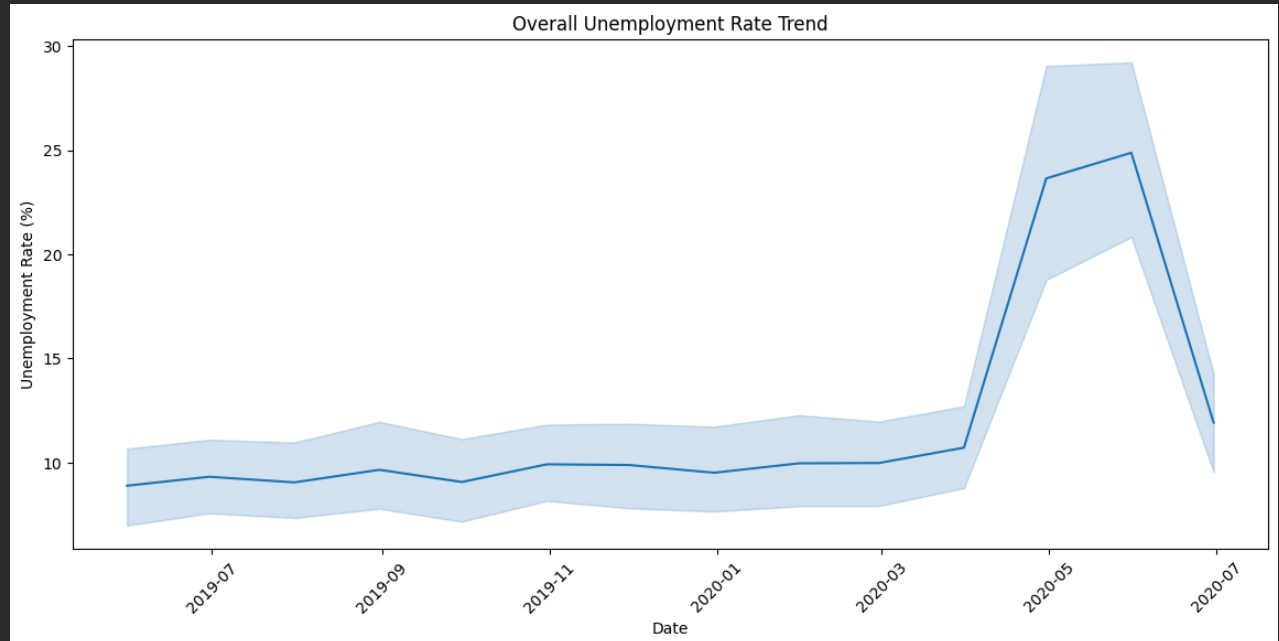
```
plt.figure(figsize=(14,6))
```

```
sns.lineplot(
    data=df,
    x='Date',
    y='Estimated Unemployment Rate (%)'
)
```

```
plt.title("Overall Unemployment Rate Trend")
plt.xlabel("Date")
plt.ylabel("Unemployment Rate (%)")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



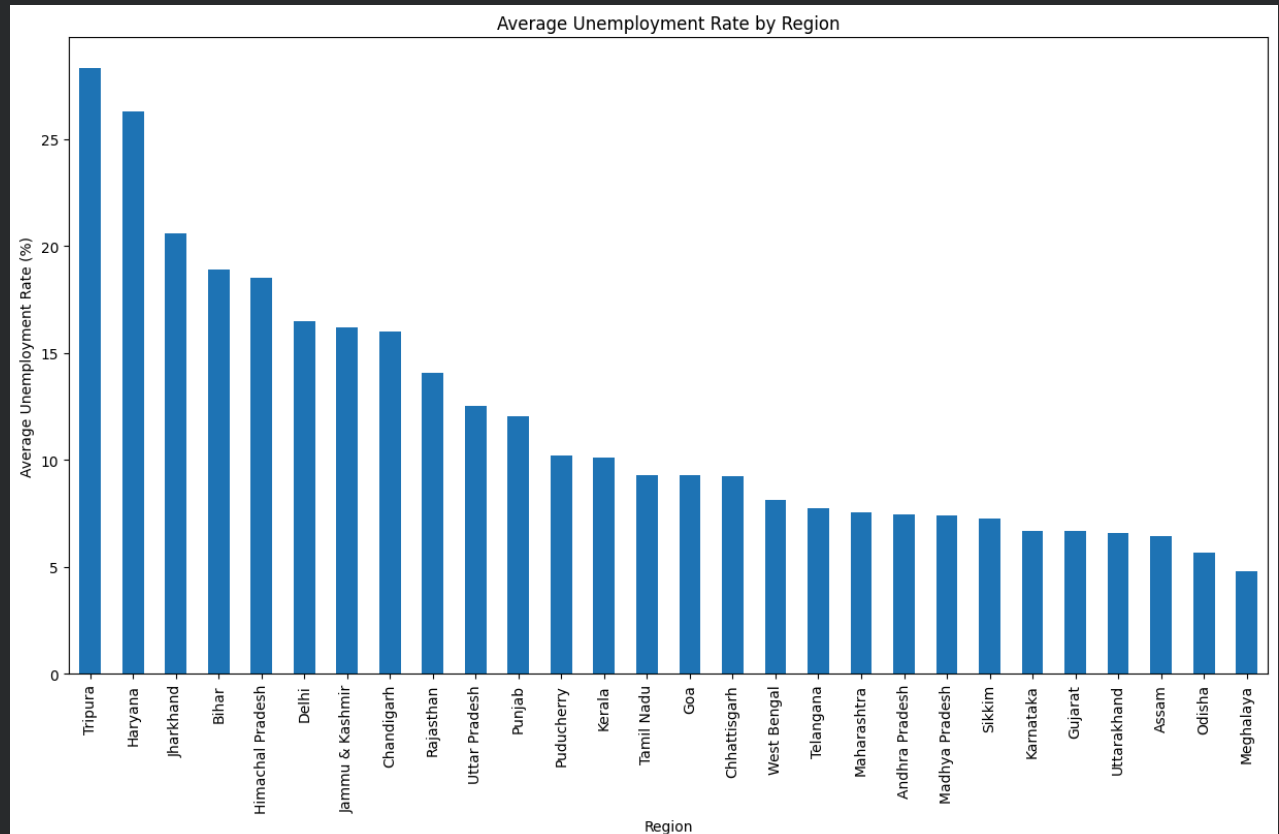
```
# Regional Unemployment Analysis
plt.figure(figsize=(15,8))

top_regions = df.groupby('Region')['Estimated Unemployment Rate (%)'].mean().sort_values(ascending=False)

top_regions.plot(kind='bar')

plt.title("Average Unemployment Rate by Region")
plt.xlabel("Region")
plt.ylabel("Average Unemployment Rate (%)")

plt.show()
```



## Impact of COVID-19 on Unemployment

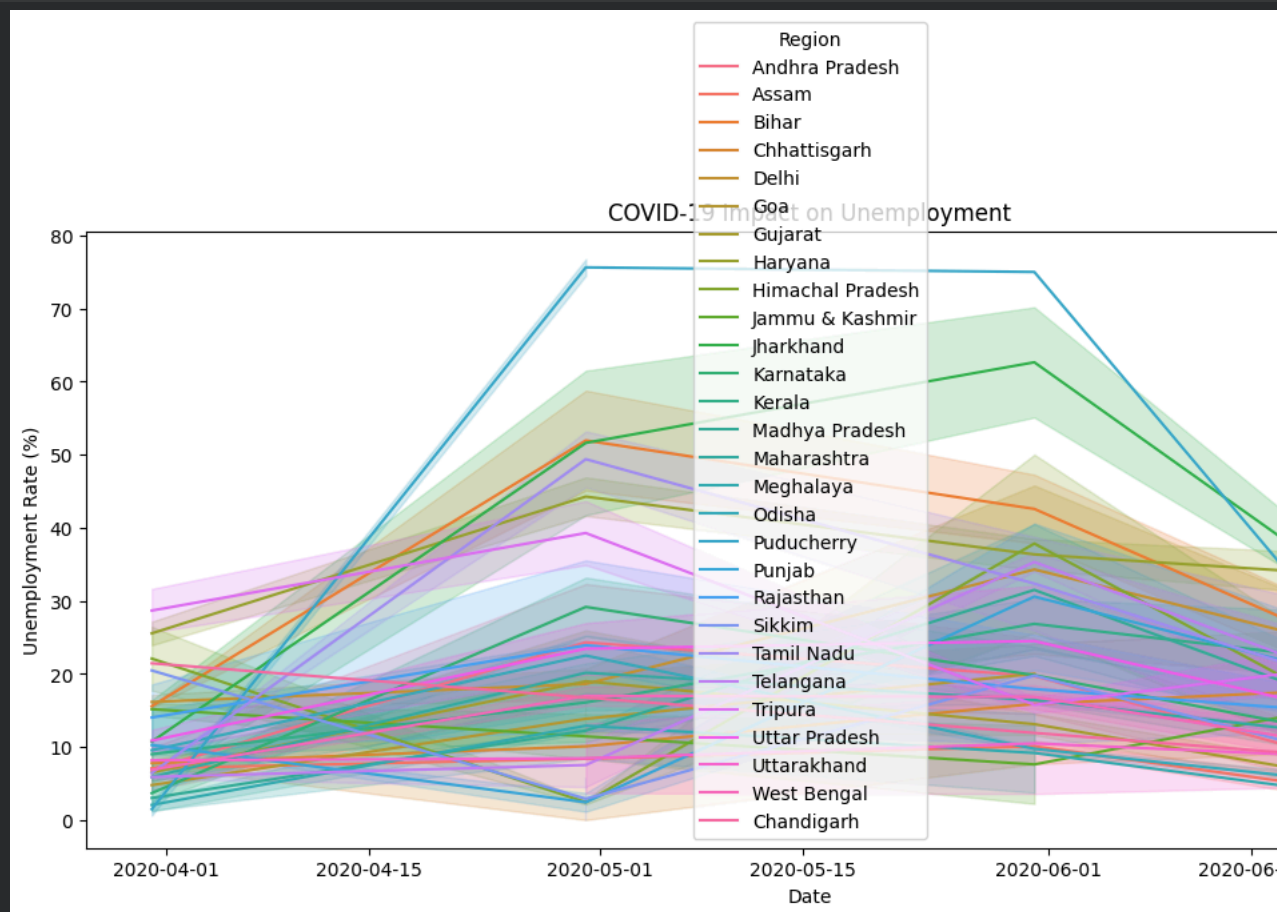
```
covid_period = df[df['Date'] >= '2020-03-01']

plt.figure(figsize=(14,6))

sns.lineplot(
    x='Date',
    y='Estimated Unemployment Rate (%)',
    hue='Region',
    data=covid_period
)

plt.title("COVID-19 Impact on Unemployment")
plt.xlabel("Date")
plt.ylabel("Unemployment Rate (%)")

plt.show()
```



## Monthly Trend Analysis (Seasonality)

```
df['Month'] = df['Date'].dt.month

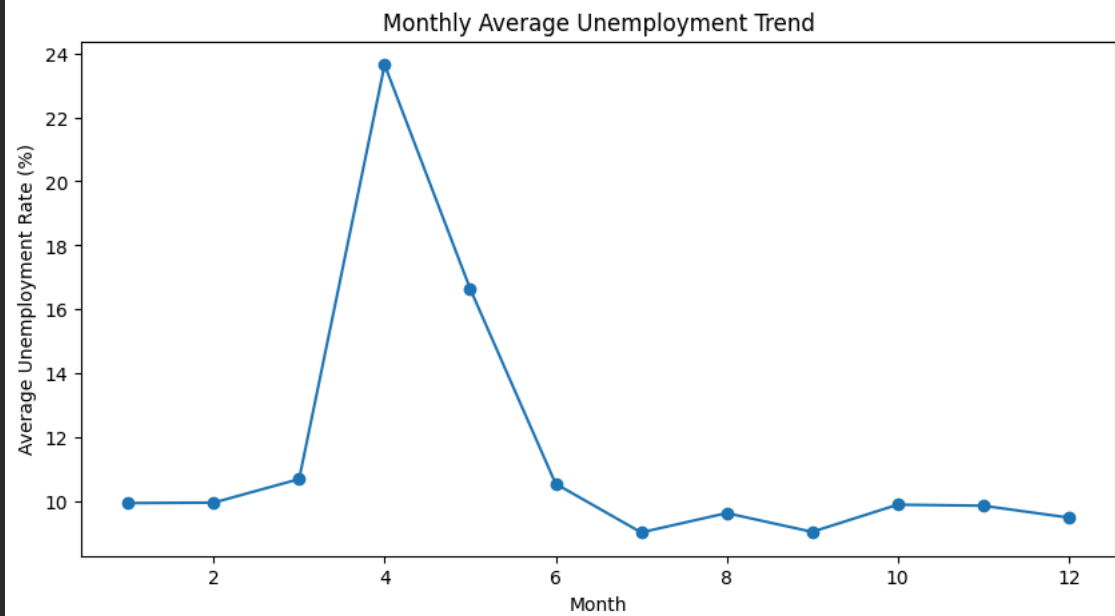
monthly_avg = df.groupby('Month')['Estimated Unemployment Rate (%)'].mean()

plt.figure(figsize=(10,5))

monthly_avg.plot(marker='o')

plt.title("Monthly Average Unemployment Trend")
plt.xlabel("Month")
plt.ylabel("Average Unemployment Rate (%)")

plt.show()
```



- Observation
- Seasonal variations may occur due to:
- Agricultural cycles
- Festivals
- Tourism
- Economic slowdowns

```
# Correlation Analysis
correlation = df.corr(numeric_only=True)

plt.figure(figsize=(8,5))

sns.heatmap(correlation, annot=True, cmap='coolwarm')

plt.title("Correlation Matrix")

plt.show()
```

